chamber causes condensation along its path—the well-known cloud tracks.

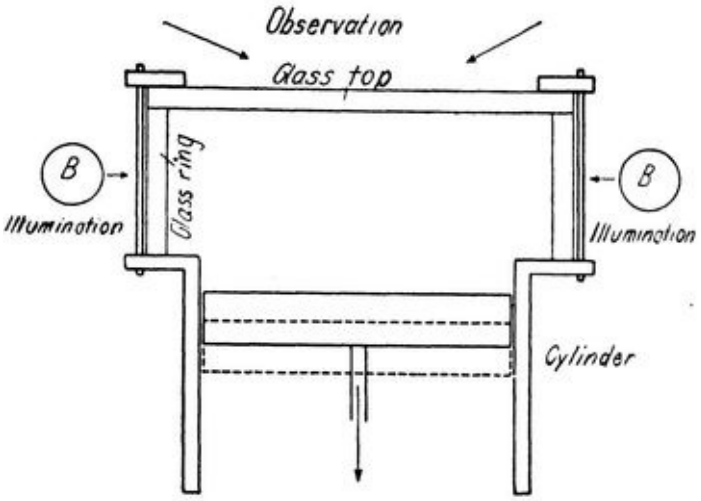


Figure 32.—A simplified sketch of Wilson's cloud chamber.

Many phenomena which one would like to observe in the cloud chamber, in particular the phenomena of cosmic radiation, are extremely rare, and the observer may have to wait for several hours before they eventually take place. The chances of encountering just such a reaction when observing an expansion, are very small indeed. Therefore, if we had to rely on chance alone for the eventual observation of such rare occurrences, studies of this kind would require a very great deal of time. However, this handicap is eliminated by skilfully connecting the cloud chamber with a counter which acts, so to speak, as a sentry at the gate of the cloud chamber. The counter is adjusted so that it will react to the specific phenomenon which it is desired to observe in the cloud chamber. If such a phenomenon actually takes place, the counter immediately effects the expansion through an amplifier. This takes place so fast that the ions formed in the chamber have not yet diffused away from the paths of the particles, and they are therefore visible as cloud tracks. This is the method which has yielded the most important data known concerning the nature of cosmic radiation, during the past decade.

Finally, the photographic plate, too, can be used as a detector of charged particles. An example has been shown already in Figure 25.

These methods enable us to count or detect every radiation with which an electric charge is associated (alpha and beta radiation, as well as all other types of nuclear debris which carry an electric charge), and also gamma-ray photons, so that the question of the detection of neutrons remains the only one still to be explained. Since neutrons carry no electric charge, they do not themselves produce ionization, and therefore we must rely on the observation of a secondary effect in order to detect their presence. The simplest device for this purpose is a boron counter. The inside of the wall of this counter is lined